

Nassau Candy Distributor: Profitability & Cost Structure Analysis

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1. Executive Summary

This report presents a comprehensive profitability and cost structure diagnostic for Nassau Candy Distributor. The objective of this analysis was to identify profit concentration risks, evaluate product-level margins, and detect pricing inefficiencies across the catalog. The analysis transitioned from initial static Excel PivotTables into a fully interactive Python Streamlit dashboard. The core finding reveals a severe over-dependency risk: just four products account for nearly **80%** of total company profit, while a "long tail" of low-margin products actively drains logistical resources.

2. Exploratory Data Analysis (EDA) Overview

The dataset consisted of order-level transactional data, including Sales, Costs, Units Sold, and Gross Profit across multiple product divisions (Chocolate, Sugar, and Other).

- Data aggregation was performed at the Product Name level to establish baseline Key Performance Indicators (KPIs): Gross Margin (%), Profit per Unit, and Total Profit Contribution.
- Cost vs. Sales scatter mapping was utilized to visually group products into four actionable quadrants: Cash Cows, Renegotiation Targets, Repricing Candidates, and Discontinuation Risks.

3. Key Insights & Findings

- **The 80/20 Profit Concentration (Pareto Rule):** The business is highly top-heavy. The top four products (Wonka Bar - Scrumdiddlyumptious, Triple Dazzle Caramel, Milk Chocolate, and Nutty Crunch Surprise) generate **77.30%** of the entire gross profit.
- **Margin Vulnerabilities:** Several products operate on unsustainably low margins. Specifically, "Fun Dip" operates at a **13.14%** margin, and "Kazookles" operates at a

12.30% margin, meaning any minor supply chain disruption will cause these products to lose money.

- **Cost Inefficiencies:** The product "Lickable Wallpaper" presents a unique financial trap. Despite a high percentage margin on paper, the absolute raw cost to generate its revenue is disproportionately high (\$3,930 in costs to generate \$1,429 in sales).
- **Division Performance:** The Chocolate division is the financial anchor of the company, generating the vast majority of revenue while maintaining a highly healthy average margin of **67.45%**.

4. Strategic Recommendations

Based on the data diagnostics, the following actions are recommended to optimize the product portfolio:

1. **Discontinuation Review:** Immediately review "Fun Dip" and "Kazookles" for discontinuation. They are tying up warehouse space and cash flow for practically zero bottom-line return.
2. **Cost Renegotiation:** Urgently renegotiate supplier costs for "Lickable Wallpaper" to reduce the capital required to stock it.
3. **Repricing Strategy:** Execute a slight price increase on mid-tier candies like "SweeTARTS", "Laffy Taffy", and "Nerds." Their sales volumes are low enough that a price hike will not severely impact overall company revenue, but it will improve their individual margins.
4. **Protect the Core:** Secure the supply chain for the top 4 Wonka Bars. Because the company is heavily dependent on them, any shortage of ingredients (like cocoa) for these specific bars poses a severe risk to the entire business.